

9. Linear vs Polynomial Regression

Using the **House Size vs House Price** dataset

```
from sklearn.linear_model import LinearRegression
```

```
from sklearn.preprocessing import PolynomialFeatures
```

```
X = [[800],[1000],[1200],[1400],[1600],  
     [1800],[2000],[2200],[2400],[2600]]
```

```
y = [32,40,52,68,88,112,140,172,208,248]
```

```
# Linear
```

```
linear = LinearRegression()
```

```
linear.fit(X,y)
```

```
lp = linear.predict([[1700]])
```

```
# Polynomial
```

```
poly = PolynomialFeatures(degree=2)
```

```
Xp = poly.fit_transform(X)
```

```
model = LinearRegression()
```

```
model.fit(Xp,y)
```

```
pp = model.predict(poly.transform([[1700]]))
```

```
print("Linear Prediction:", round(lp[0],2))
```

```
print("Polynomial Prediction:", round(pp[0],2))
```

OUTPUT

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

= RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB/machine learning lab/EXP 7.py
Linear Prediction: 116.0
Polynomial Prediction: 99.5